A Null-Peripheral Subcase of Weak-Star Limit Decay in Shift-Invariant Preduals of $\ell_1(\mathbb{Z})$

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2026-06-29

Status. Candidate partial result, likely valid.

Source Question

The source paper is Daws, Haydon, Schlumprecht, and White, *Shift invariant preduals of* $\ell_1(\mathbb{Z})$, arXiv:1101.5696 [1]. Its Question 5 asks:

> Characterise those $a \in \ell_1(\mathbb{Z})$ which occur as weak*-limit points of $\{\delta_n : n \in \mathbb{Z}\}$. In particular, is the condition $\lim_n \|a^n\|_\infty = 0$ necessary as well as sufficient?

The question appears on PDF page 29:

6 Questions

We end the paper with a range of open questions regarding these preduals.

1. Describe all possible semigroups $\mathcal{S}$ arising as part of a minimal pair inducing a shift-invariant predual of $\ell_1(\mathbb{Z})$.
2. What are the Banach space isomorphism classes of shift-invariant preduals?
3. What is the Banach space isomorphism class of the shift-invariant predual constructed in Theorem 5.8?
4. For any countable ordinal α, does there exist a shift-invariant predual with Szlenk index at least α?
5. Characterise those $a \in \ell_1(\mathbb{Z})$ which occur as weak*-limit points of $\{\delta_n : n \in \mathbb{Z}\}$. In particular, is the condition $\lim_n \|a^n\|_\infty = 0$ necessary as well as sufficient?
6. The concrete shift-invariant preduals $F^{(\lambda)}$ of Section 3 are *cyclic* in that they are the minimal closed, shift-invariant subspaces containing the specified element x_0. Characterise the cyclic shift-invariant preduals of $\ell_1(\mathbb{Z})$.

Result

For $a \in \ell_1(\mathbb{Z})$, write

$$\widehat{a}(z) = \sum_{k \in \mathbb{Z}} a(k) z^k, \qquad z \in \mathbb{T}.$$

The convolution power a^n has Fourier transform $\widehat{a}^{\,n}$.

Theorem 1 (Null peripheral criterion)**.** *Let $E \subseteq \ell_\infty(\mathbb{Z})$ be a shift-invariant predual of $\ell_1(\mathbb{Z})$. Suppose $a \in \ell_1(\mathbb{Z})$ is a weak*-limit point of $\{\delta_n : n \in \mathbb{Z}\}$ for this predual. If*

$$m\big(\{z \in \mathbb{T} : |\widehat{a}(z)| = 1\}\big) = 0,$$

where m is normalized Haar measure on $\mathbb{T}$, then

$$\|a^n\|_\infty \longrightarrow 0.$$

Corollary 1 (Finitely supported candidates)**.** *The necessity direction in Question 5 is affirmative for every finitely supported weak*-limit candidate a which is not a unimodular translate of a point mass $\lambda\delta_k$. In particular, if a finitely supported a actually occurs as a weak*-limit point of point masses in a shift-invariant predual, then $\|a^n\|_\infty \to 0$.*

Proof Intuition

The source semigroup machinery already gives the key algebraic fact: every weak*-limit point a of point masses is power-bounded for convolution. On the Fourier side this says that the powers $\widehat{a}^{\,n}$ stay bounded in the Wiener algebra, hence $|\widehat{a}| \le 1$. Once the set where $|\widehat{a}| = 1$ has measure zero, dominated convergence makes $\widehat{a}^{\,n} \to 0$ in $L^2(\mathbb{T})$. Plancherel then controls every Fourier coefficient uniformly, which is exactly the ℓ_∞-norm of the convolution power. The finite-support corollary follows because a non-monomial Laurent polynomial cannot have modulus one on a positive-measure subset of the circle.

Proof

We first record the source-paper consequence used here. By Proposition 4.3 of [1], every shift-invariant predual E is induced by a compact semitopological semigroup $\mathcal{S}$ and a bounded Banach-algebra homomorphism

$$\Theta : M(\mathcal{S}) \to \ell_1(\mathbb{Z})$$

which is a projection onto the copy of $\ell_1(\mathbb{Z})$. Proposition 4.4 of [1] identifies the weak*-closure of the point masses as

$$\{\Theta(\delta_\gamma) : \gamma \in \overline{\mathbb{Z}}^{\mathcal{S}}\}.$$

Thus, if a is a weak*-limit point of point masses, then $a = \Theta(\delta_\gamma)$ for some $\gamma \in \mathcal{S}$. Since Θ is an algebra homomorphism,

$$a^n = \Theta(\delta_\gamma)^n = \Theta(\delta_{n\gamma}),$$

and therefore

$$\sup_{n\ge1} \|a^n\|_1 \le \|\Theta\| < \infty.$$

Let $f = \widehat{a} \in A(\mathbb{T})$, the Wiener algebra. From the preceding bound,

$$|f(z)|^n = |\widehat{a^n}(z)| \le \|a^n\|_1 \le \|\Theta\| \qquad (z \in \mathbb{T},\ n \ge 1),$$

so $|f(z)| \le 1$ for all $z \in \mathbb{T}$. Assume now that $m(\{|f| = 1\}) = 0$. Then $|f(z)|^{2n} \to 0$ for almost every $z \in \mathbb{T}$, and the functions are dominated by 1. Dominated convergence gives

$$\int_{\mathbb{T}} |f(z)|^{2n}\, dm(z) \longrightarrow 0.$$

By Plancherel,

$$\|a^n\|_\infty \le \|a^n\|_2 = \left(\int_{\mathbb{T}} |f(z)|^{2n}\, dm(z)\right)^{1/2} \longrightarrow 0.$$

This proves Theorem 1.

For the finite-support corollary, $\widehat{a}$ is a Laurent polynomial. If $\widehat{a}$ is not of the form λz^k and the set $\{|\widehat{a}| = 1\}$ had positive measure, then the real-analytic function $|\widehat{a}(e^{it})|^2 - 1$ would vanish on a set with an accumulation point, hence would vanish identically. Thus $|\widehat{a}| = 1$ on all of $\mathbb{T}$. Writing $\widehat{a}(z) = z^{-N}p(z)$, with p a polynomial, we get $|p| = 1$ on $\mathbb{T}$. A polynomial inner function is a unimodular monomial, so $\widehat{a}(z) = \lambda z^k$, contrary to the assumption. Hence the peripheral set has Haar measure zero, and the theorem applies.

Finally, Proposition 4.6 of [1] rules out $\lambda\delta_0$ as a weak*-limit point; by shift-invariance, the same is true for every unimodular translate $\lambda\delta_k$. Thus every finitely supported weak*-limit point falls under the non-monomial case above.

Limitations and Novelty Check

This packet does not solve Question 5 in full. The remaining case is a power-bounded element $a \in \ell_1(\mathbb{Z})$ whose Fourier transform has unit modulus on a positive-measure subset of $\mathbb{T}$, while a is not a unimodular point mass. The previous attempt in this run isolated power-boundedness but did not close even the null-peripheral case.

Local run indexes were searched for arXiv:1101.5696 and core terms including "shift invariant preduals", "weak star", "ell_1(Z)", and "power-bounded". The only direct run hit was the earlier unfinished attempt `1101.5696_weakstar_limit_power_decay.md`; no solution packet for this subcase was present. A bounded web search for power-bounded Wiener-algebra and convolution-power concentration keywords did not identify an exact full answer during this pass.

Human Review Notes

The main proof is the Plancherel estimate in Theorem 1. Reviewers should check the source citations for the inducing-pair framework and the final shift-invariance use excluding $\lambda\delta_k$. The finite support corollary uses only the standard fact that a polynomial inner function is a unimodular monomial.

References

[1] Matthew Daws, Richard Haydon, Thomas Schlumprecht, and Stuart White, *Shift invariant preduals of* $\ell_1(\mathbb{Z})$, arXiv:1101.5696.